Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

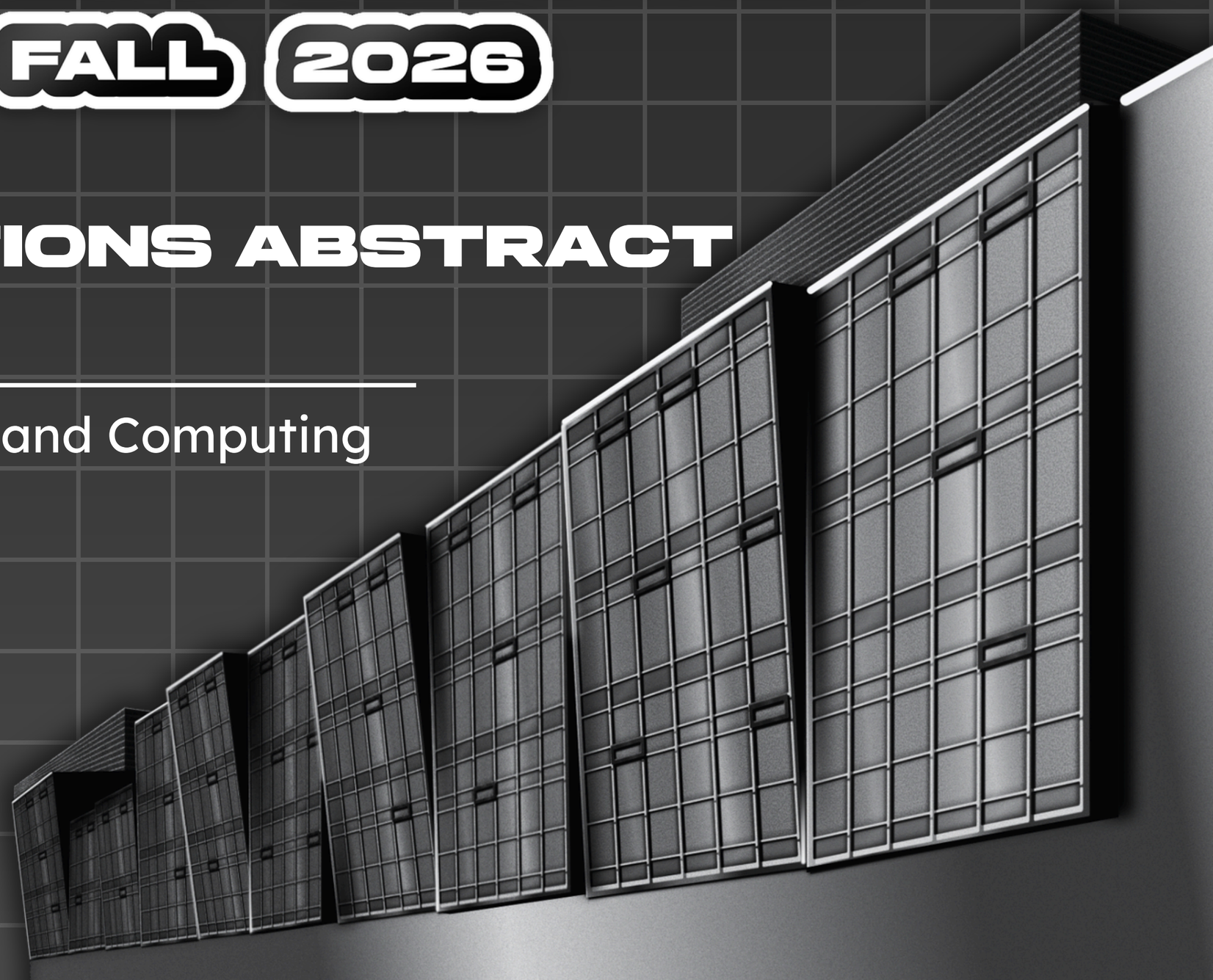
TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

FALL 2026

COMMUNICATIONS ABSTRACT

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre



ENGINEERING COMMUNICATION

Participant Competition Package

Theme: Feeding the Future: Designing Smart Infrastructure for Urban Food Security

Competition Date: Sunday, November 8, 2026

Submission Deadline: Friday, November 6, 2026, at 11:59 PM Eastern Time.

Team Size: 2 competitors

Competition Format: Prepared Public-Information Presentation

Presentation Duration: Maximum 20 minutes

Q&A: 5 minutes

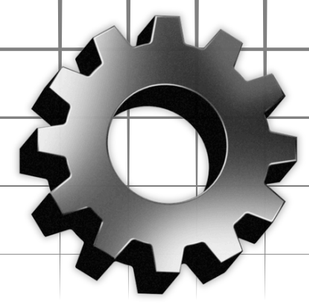
1. COMPETITION OVERVIEW

Engineering projects can improve communities, but their success depends partly on whether the public understands how they work, why they are needed, and what impacts they may have.

Engineers must communicate technical information to people who may not have engineering backgrounds. This includes explaining complex systems clearly, addressing public concerns, and presenting project benefits and limitations accurately.

In the MEC 2026 Engineering Communication Competition, teams will act as engineering communication representatives responsible for presenting an approved infrastructure project to members of the Toronto community.

The project has already been designed and approved within the fictional competition scenario.



Your responsibility is to communicate the project, not redesign it.

Teams will receive the complete project information approximately six days before the competition and use that preparation period to develop their presentations.

On competition day, teams will present to judges acting as community stakeholders and respond to questions concerning the project.

2. COMPETITION SCENARIO

Project: Toronto Community Food Access and Smart Distribution Network

2.1 Background

Toronto faces challenges related to food affordability, access to fresh produce, and the distribution of perishable food products.

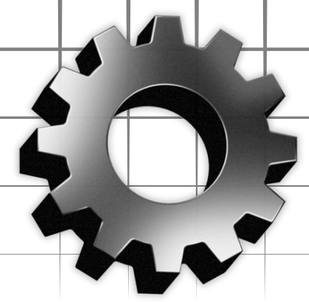
In some neighbourhoods, residents must travel considerable distances to access affordable fresh food. At the same time, food-distribution organizations must manage transportation costs, product spoilage, refrigeration requirements, and unpredictable demand.

To address these challenges, the fictional **Toronto Community Food Infrastructure Initiative (TCFII)** has approved a pilot project that combines refrigerated community food-distribution stations with centralized monitoring and coordinated delivery operations.

The project is intended to improve the accessibility, reliability, and efficiency of fresh-food distribution.

The engineering design has already been approved for the purposes of this competition.

The public is now being introduced to the project before its implementation.



2.2 Your Assignment

Your team has been selected to explain the approved project to residents, local businesses, community organizations, and other stakeholders.

Some audience members may support the project, while others may be concerned about its cost, environmental impact, accessibility, maintenance, reliability, or long-term operation.

Your presentation must help these stakeholders understand the engineering system and make informed judgments about its expected community impacts.

3. APPROVED ENGINEERING PROJECT

The proposed pilot consists of three refrigerated food-distribution stations installed at existing community facilities.

Each station will include refrigerated storage, temperature monitoring, a digital monitoring connection, and a community pickup interface.

Food suppliers and participating organizations will coordinate scheduled deliveries to the stations.

3.1 Project Specifications:

The following specifications are fixed for the fictional competition scenario.

Number of pilot stations	3
Refrigerated storage capacity	500 L per station
Total refrigerated capacity	1,500 L
Normal refrigerated operating range	2°C to 5°C
Temperature monitoring	Continuous sensor monitoring
Data reporting interval	Every 5 minutes
Temperature alert condition	Reading outside 2°C–5°C
Monitoring platform	Centralized digital dashboard
Station access	Designated community pickup hours
Primary electrical supply	Municipal electrical grid
Backup capability	Monitoring and alert backup power only
Planned operating life	10 years, subject to maintenance

Important Operational Limitation

The approved backup power system supports temperature monitoring and alert functions only. It does not provide backup refrigeration during a power outage.

During an electrical disruption, the stations may be unable to maintain their normal refrigerated operating range of 2°C to 5°C. Competitors should explain this limitation accurately and discuss its implications for food safety, operational reliability, and community confidence.

Teams are not expected to redesign the approved backup system but should demonstrate an understanding of the associated risks and safeguards.

3.2 Approved Project Budget

Category	Cost (CAD)
Refrigerated stations	$180,000
Sensors and monitoring hardware	$45,000
Software and integration	$55,000
Site preparation and installation	$75,000
Accessibility features	$30,000
Testing and training	$25,000
Contingency	$40,000
Total Capital Budget	**$450,000**

Estimated Annual Operating Cost: $75,000 CAD

Implementation Timeline

The approved pilot implementation period is 12 months.

- **Months 1–2:** Site coordination and planning
- **Months 3–5:** Procurement and preparation

- **Months 6–8:** Installation and integration
- **Months 9–10:** Testing and training
- **Months 11–12:** Pilot launch and evaluation

4. COMPETITION TASK

Your team must prepare a public-information presentation explaining the approved engineering project to a non-technical audience.

Your presentation should demonstrate an understanding of the project's purpose, engineering operation, implementation, costs, expected impacts, and relevant risks or limitations.

Teams are responsible for determining how to organize and communicate the information effectively.

Competitors may conduct supporting research to strengthen their understanding of the engineering concepts involved and provide relevant context for their presentation. However, the approved project specifications, budget, and implementation timeline must remain unchanged.

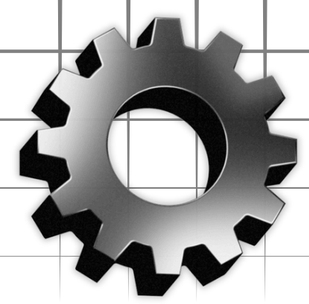
Teams are expected to explain the benefits, limitations, risks, and potential community impacts of the approved project rather than propose a different engineering solution. Competitors may discuss reasonable concerns or recommendations for future improvement, provided these are clearly distinguished from the approved project.

5. PRESENTATION REQUIREMENTS

Each team must deliver:

Required Pre-Competition Submissions

Each team must submit the following documents through the official Engineering Communication Google Form:



1. Presentation Abstract

A written summary of the approved engineering project and how the team intends to communicate its purpose, operation, impacts, and key considerations to a non-technical audience.

The Presentation Abstract is a mandatory pre-competition submission used to provide organizers and judges with an overview of the team's planned presentation. It does not receive a separate numerical score. The presentation and community Q&A will be evaluated using the official Engineering Communications scorecard.

- Maximum 200 words
- Maximum 1 page
- 12-point font
- 1-inch margins
- 1.5 line spacing
- Include team name and both competitors' names
- Submit as a PDF

2. Presentation Slides

A PDF copy of the team's prepared public-information presentation.

- Include team name and both competitors' names
- Submit as a PDF
- Cite external sources appropriately

Both the Presentation Abstract and Presentation Slides must be submitted by Friday, November 6, 2026, at 11:59 PM Eastern Time, through the official submission form provided in the pre-competition email.

Expected Presentation Content

The presentation should communicate the following information in a clear and accessible manner:

1. Community Problem: The food-accessibility and distribution challenges the approved project is intended to address.
2. Project Overview: The purpose and main components of the Toronto Community Food Access and Smart Distribution Network.
3. System Operation: How the refrigerated stations, temperature sensors, monitoring platform, and community pickup process work together.
4. Budget and Implementation: The approved capital budget, estimated annual operating cost, and 12-month implementation timeline.
5. Community Impacts: The expected social, environmental, and economic implications of the project.
6. Risks and Limitations: Relevant concerns involving food safety, power outages, maintenance, accessibility, reliability, and operational trade-offs.

Teams should use appropriate visuals and plain language to help a non-technical audience understand the project. Technical accuracy, balanced explanations, and effective communication are essential.

Competition-Day Requirements

Prepared Presentation: Each team will deliver a public-information presentation lasting a maximum of 20 minutes, using the presentation slides submitted before the competition.

Community Q&A: Following the presentation, teams will participate in a 5-minute question-and-answer session with judges acting as non-technical community stakeholders.

Both competitors are expected to participate meaningfully in the presentation and demonstrate an understanding of the approved engineering project.

6. COMPETITION RULES

1. Teams must communicate the approved engineering project provided by MEC organizers. Proposing a replacement engineering solution is not permitted.
2. Teams may use supporting research to understand technical concepts but must distinguish external information from the approved project specifications.
3. Any external sources used in the presentation must be appropriately cited.
4. Competitors must not fabricate project specifications, costs, implementation details, or performance results. Any expected benefits or potential outcomes discussed beyond the approved project information must be clearly identified as estimates, assumptions, or possibilities rather than verified project results.
5. AI-assisted preparation is permitted only in accordance with MEC's applicable AI policy. Teams remain responsible for the accuracy and originality of their work.
6. Presentations must remain professional, respectful, and appropriate for a non-technical public audience.
7. Teams must follow all competition-day instructions and submission requirements communicated by MEC organizers.

7. JUDGING CRITERIA

Category	Points
Technical Accuracy & Project Understanding	20
Communication & Audience Awareness	20
Impacts, Risks & Trade-offs	20
Presentation & Visual Communication	20
Community Q&A	20
Total	**100**